

BOLT

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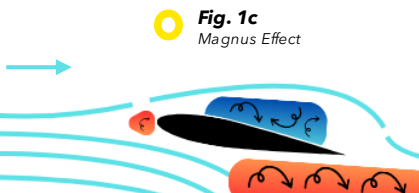
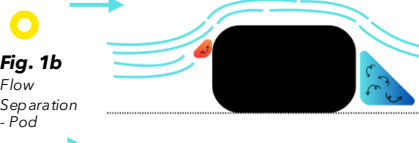
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STEM RACING ENGINEERING PORTFOLIO

Energy: Everything We need

Energy defines how **physical interactions** occur, shaping everything from **motion to heat**. In a car, fuel energy transforms into the rotational motion of the wheels, producing speed. However, not all energy contributes to motion – some is lost as heat due to friction and air resistance, explaining why cars slow down. The **law of conservation of energy** states that energy **cannot be created or destroyed**, only transformed between forms like kinetic and thermal energy. This interchange shows how all motion and mechanical processes, including racing, are governed by the flow and conversion of energy. So, for the purpose of simplicity and physical accuracy, we tend to **generalise values under the common langue of energy**, to easily compare them and apply them to equations – energy is a far purer value compared to that of force.

Adverse Pressure Regions:



Flow Separation:
Flow separation occurs on a relatively moving body. When air hits a face, it is forced around the object, but it **travels slower** than before upon contact – this **creates high pressure**. When area leaves the surface of the body, a gap is formed behind it which the airflow cannot immediately fill: it is not in the direction of the flow's momentum. **This creates low pressure behind it**, also known as the **wake region** of the object. Flow separation can occur on any shape but is typically associated with the wheels and pods.

Magnus Effect:
The magnus effect occurs on the **front wheel** of our car, and results in the formation of a **pressure gradient**. It occurs on a rotating wheel experiencing flow separation (see previous) creating a low-pressure zone behind it and high pressure in front. **In order to equalize pressure**, the high-pressure area pushes the wheel while the semi-vacuum pulls, causing the wheel to experience backwards force, or drag.

Bernoulli Effect:
While common in airplane discussion, the Bernoulli effect affects the normal force of the car: when airflow hits an **aerofoil at an angle**, most of it is **pushed in the aerofoil's angle**, creating high pressure under the wing. The high pressure pushes the wing away from it, creating lift (+ or -).

Stability:

Centre Of Mass Offset (CMO):
CMO refers to the **distance between the ground and the centre** of mass. A lower CMO will result in a more stable car because other forces will exert most energy on the CM: Higher CM means forces are larger higher up, increasing turn-over and yaw potential

Wheelbase:
Wheelbase refers to the 2D area between the 4 wheels. **A larger wheelbase will result in a more stable car**, as the mass of the car is spread out over a larger area of support. For this reason, we sought to have the largest possible wheelbase without **sacrificing weight, while maintaining legality**.

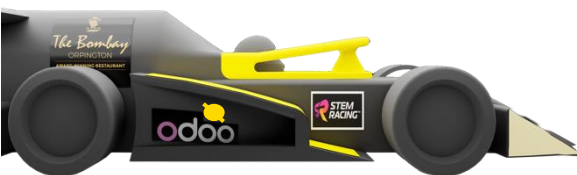


Fig. 1e
COM location Final Car

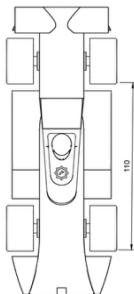
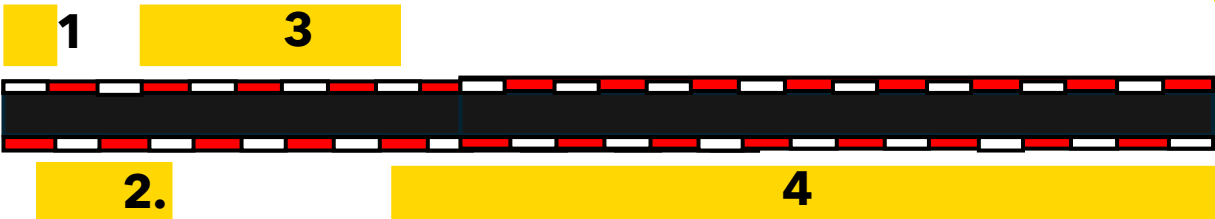


Fig. 1f
Car Wheelbase (110mm)

The 4 Stages of Race:

Fig. 1g
Track Diagram



1. Launch:
When the gas canister is triggered and released, the **333 joules** of energy locked away in the CO2's pressure is released. This is by far where majority of energy loss occurs due to turbulence in the canister and loss in gas expansion, but without LERs this phase is inevitable and **cannot be controlled**.

3. Acceleration:
In the acceleration phase, the car is gaining speed- this is not a drag period, but is, of course, still affected by the aerodynamic forces explained in the drifting phase. To a large extent, these are amplified due to the fact **that aerodynamic forces scale exponentially with speed**, defined as V squared in the aerodynamic equation. Thus, this phase can be affected, as the less opposing forces acting on the car, **the longer time it will accelerate**, and the faster our time will be.

4. The Drifting Phase:
This phase is the largest and **most critical part of the race**, as it involves factors we can control. Unlike the fixed energy loss in the gas canister, here we **can minimize energy** loss through car optimization. Since the car no engine, and cannot accelerate, therefore depending entirely on aerodynamics, performance is determined by **4 key energy strippers**:

Drag.
Drag is perhaps the **most recognizable aspect of motorsports engineering**, but it's not as dominant as we once believed, as the ML sim proved. Nevertheless, **drag is not negligible**; it still plays a crucial role in securing lap times down to the hundredths of a second. Put simply, **drag is the friction between the car and the air**. As airflow encounters the car's surface, some of the car's kinetic energy is converted into wasted heat. This happens because, according to the principle of **energy conservation**, any force applied to an object must result in an energy transfer: as air flows over the car, repeated interactions between the car's surface and the air molecules generate heat, slightly slowing the car in the process. The equation for drag is as follows:

$$\frac{1}{2} \rho u^2 C_d A d$$

Where ρ is density, u is air speed relative to car, C_d is the drag Co., A is the frontal cross section and d is the track distance. The only variable in our control is the drag co-efficient, which can be optimised through testing.

Downforce. Downforce plays a key role in the equation for rolling resistance, as it influences the normal force. The normal force of an object refers to the force acting on our car perpendicular to the direction of travel, or the up and down. **Normal force, when the car is in motion, is a combination of downforce and weight**. By lowering the normal force, we lower the rolling resistance, as the equation suggests. To reduce the normal force, we can produce **small amounts of lift to balance the weight of the car**: as long as it **remains above 0**, the car will remain on track.

2. Tipping Moment
While doing research, we explored **slow-motion videos of old F1 in school's car launches** to understand the **complex physics of this moment**. We noticed a common trend: immediately after launch, the cars tended to 'tip' onto their front wing, raising the rear wheels of the track (as you can see in fig. 1h). We realized that this would result in major energy loss, as part of the **canister's thrust pushed the car down**, rather than straight, wasting the thrust. Furthermore, it provided the risk that the front wing could scrape against the track, producing a high amount of friction and potentially damaging the wing.



Fig. 1h
Tipping Movement (credit: Team Aero flux)

In order to prevent this, we needn't distribute mass evenly across car, **rather, have the center of mass** (the point at which the normal force is the highest) **towards the rear of the car**. By doing so, we cancel out the forces of thrust pushing the car down by gravity, essentially turning the car into a lever, as fig. 3 suggests.

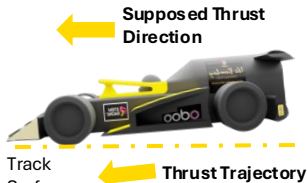
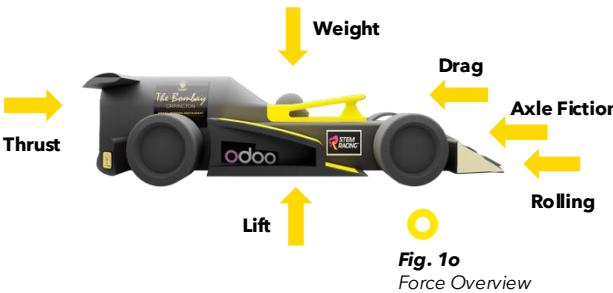


Fig. 1i
Tipping Movement

Overview Of Force:

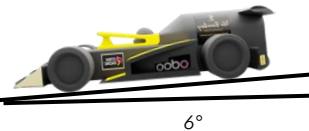


Rolling Resistance.
Rolling resistance relates to the wheels: it refers to the **energy lost due to friction between the wheels and the track**. Friction is indeed, also a result of the conservation of energy, mentioned before. The equation for it is as follows:

$$C_r W F_d = F_r$$

Where C_r = the rolling resistance co-efficient, W = weight, F_d = downforce and F_r = rolling resistance force. **By running a simulation**, we discovered that the rolling resistance coefficient can be calculated via this experiment:

Fig. 1p
Rolling resistance Experiment



We found out the **lowest angle of incline** at which the car does not move (due to gravity on a slope) was **6°**. Our simulation told us that the rolling resistance co-efficient was **equal to tan(6)**, which is 0.105, and cannot be changed through design optimisation.

Axle Friction.
Axle friction, as the name suggests, **involves the heat energy lost from the friction between the surface of the axles and the axle bushes**(upon which they rest). Unlike the other factors, the car design cannot influence the energy loss that occurs here because the dev axle system is out of our control. However, we can reduce the axle friction through application of lubricant:

WHAT MAKES A CAR FAST?

Development Class Changes:

Mathematical Race Model:

Prioritisation:



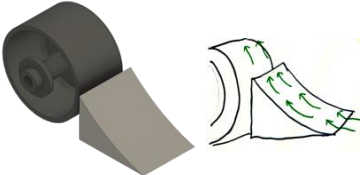
Aspect	Entry Class	Development Class
❑ Manufacture	❑ Can be done through 3D Printing or CNC	❑ Can only be done through CNC Machining
❑ Body	❑ Basic No-Go Zone	❑ More Advanced constraints
❑ Weight	❑ Minimum 65g	❑ Minimum 60g
❑ Wing Design	❑ Simpler, Limited Wing Design	❑ More Precise constraints, more scrutiny of wing design
❑ Design Freedom and Complexity	❑ Lower Complexity, Simpler shapes & Manufacturing Leniency	❑ Higher Complexity, more restrictions but higher allowed refining, strict manufacturing
❑ Clearance	❑ 1.5mm Clearance between wheels and body	❑ 3mm Clearance between wheels and body.

Fig. 2a
Entry vs Dev class

Car Components:

Front wing:

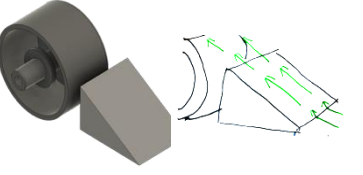
The front wing, as will be demonstrated below, has one main purpose in preventing drag: **preventing airflow from hitting the wheel**, which, rotating and being the source of movement, will severely slow our car down. However, doing so results in 3 problems: the more airflow guided up correlates to an **amplified Magnus effect**, as demonstrated in simulation. Further, it would generate much face-on drag, on the left itself, which can be resolved by a lower angle of attack. However, this, in turn, reduces the amount of airflow guided away from the wheel, **and increases weight**. Finally, a sharper angle of attack, and therefore more airflow guided up would result in **more downforce being produced**, which is unnecessary.



Curved:

Curved, is, in theory, the **best** to guide airflow over the wings, as it pushes the most airflow up.

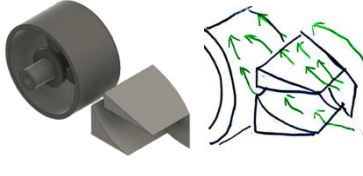
However, this does have some problems: **The Magnus effect will be amplified**, as the angle the air leaves the wheel is **higher** and more downforce is produced.



Straight:

A straight wing solves many of the problems faced by the curved design. By directing airflow at a lower angle, **the Magnus effect is reduced and less downforce is applied**.

However, more air contacts the wheels, forming more **high pressure in front**. The balance between these effects can be determined via testing.



Double:

Having 2 wings can solve some of the problems faced. It can direct air over the wheels **without increasing the angle at which that occurs, essentially the best of both worlds**.

However, it's **heavier**, and needs to be manufactured via 3d printing. Increased complexity means an increase of direct drag on the wing.



Side Shapes:

Having an which shape to use, only testing can angle on the side of the wing has 2 key benefits:

- **Reduces wing mass**
- Directs less airflow above the wheel and more around, around it. **reducing adverse pressure regions**

Ultimately, to decide tell. **Convex curve pushes the most air around**, while concave has the least face-on drag - straight is in the middle

Our **race model** allowed us to **accurately**(within the tenth of a second) predict race times, allowing us to **compare designs without any bias against characteristics**. We directly integrated it into our app for easy access without scripting. We used the following values to create it:

- ➡ Canister Energy output
- ➡ Canister Energy Power
- ➡ Rolling Resistance
- ➡ Drag
- ➡ Axle Friction

```
def track_time_simulation():
    # Initial conditions
    x = 0
    y = 0
    z = 0
    t = 0
    v_x = 0
    v_y = 0
    v_z = 0
    a_x = 0
    a_y = 0
    a_z = 0
    # Canister parameters
    E_can = 1000000
    P_can = 1000000
    # Drag parameters
    C_d = 0.5
    A_ref = 0.1
    rho = 1.2
    # Rolling resistance parameters
    C_rr = 0.01
    # Axle friction parameters
    C_af = 0.01
    # Simulation loop
    while t < 10:
        # Calculate forces
        F_drag = -0.5 * rho * C_d * A_ref * v_mag ** 2
        F_rr = -C_rr * F_weight
        F_af = -C_af * F_weight
        # Calculate acceleration
        a = (F_drag + F_rr + F_af) / m
        # Update velocity and position
        v = v + a * dt
        x = x + v_x * dt
        y = y + v_y * dt
        z = z + v_z * dt
        # Update time
        t = t + dt
    # Return final position and time
    return x, y, z, t
```

Fig. 2b
Track Time Simulation Extract

However, the model did have some large limiting factors:

- It **was assuming a perfect situation** - a purely mathematical scenario, as we did not have real results to improve it.
- It failed to account for race stability due to the **complexity** of such equations
- Virtual simulation results, which it was based off of, where no entirely accurate, and required large processing power - though this is inevitable.

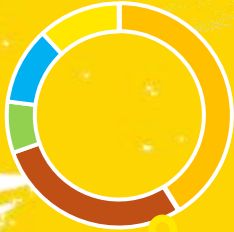


Fig. 2c
Prioritization of factors

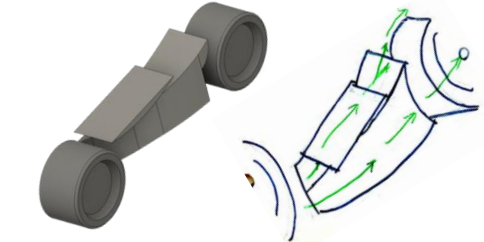
Canister Efficiency:

Canister efficiency is a **key part of our mathematical race model**, because it defines the total energy output of the canister to the kinetic energy of the car, allowing us to calculate a more accurate time. Furthermore, it aids our design process in the sense that it allows us to **accurately predict initial speeds**, and final speeds to calculate the maximum amount of lift we can generate at 20m/s(speed our simulations are run at). **This is because lift grows exponentially to speed**.

In the end, we found that the **canister was 18.6% efficient**, with the total energy locked away **being 1787J**, but the usable energy being **333J** due to **losses in gas expansion** and canister turbulence/ friction. We applied this to the canister dimensions in the race model to accurately calculate total usable energy output.

Side Pods:

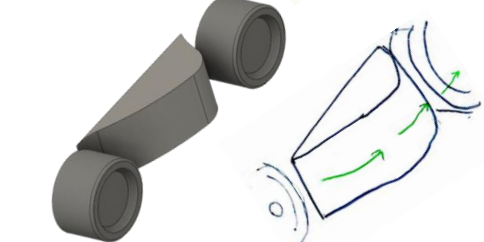
The side pods play a similar role, but for both the front and the rear wheel, thus giving it 2 purposes: To prevent the formation of a **large wheel wake** while **reducing the Magnus effect**, as well as to prevent airflow **contact with the rear wheel**. Once again, this second role does come with numerous consequences. This included **unnecessary downforce** and the rear wheel wake formation. At the same time, we found that reducing the front wheel wake came with the need to create fins, which would be **unstable and likely to break**.



Top Fin:

This was designed with the aim of **saving mass**. Rather than extruding the face below the halo platform up, this pod guides the air in its place over the rear **wheel smoothly**. The top plate also reduces the Magnus effect.

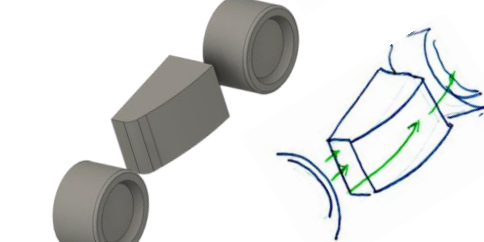
However, we were **unsure of the legality** of this, and the top plate would remain extremely fragile. **Adverse pressure gradients** may have been formed under the plane compared to outside it, creating a messy and uneven flow of air.



Curved :

Built upon last years car and inspired by previous national champion, this design aimed to punch as much airflow around the wheel rather than above it, **reducing the pressure gradients behind it and preventing airflow contact with the wheels**.

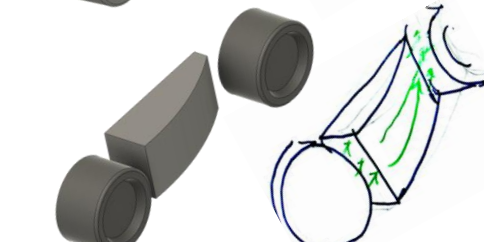
However, it does little to prevent the **low-pressure region** forming behind the wheel, as well as providing a **sharp angle of attack** which may produce **face-on drag**.



Flat-Front:

This was a tricky one. It was built upon many of the previous world champion cars, which incorporated a flat front which **previously seemed inefficient**.

However, testing would lead to the realisation that such a front would **eliminate the low pressure behind the wheel(see figure)** as well as saving mass. It also guides airflow around the wheel **instead of over. No direct disadvantages could be found**.



Reversed:

Another unusual idea. Initially, this was designed for the **sake of variety** - but actually boasted some of the best results. The flat front eliminates the **Magnus effect** on the front wheel. Mass is also minimised as it follows the shape of the no-go-zone

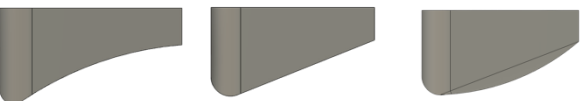
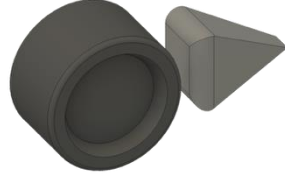
However, it **failed to guide air around the wheel**, increasing the Magnus effect on the back and **resulting in face-on wheel drag**, which was highly disadvantageous.

Rear Pods:

Simply put, the rear pods are **solely responsible for minimising rear wheel wake region**. This means that it must take on the space of the wake, to fill in the low pressure region, all done while trying to minimise mass.

Top Fin:

The rear pod has one purpose: filling up the low pressure region behind the rear wheel. This, a classic design can be **used to mimic the shape of the wheel's wake**. Different shoes may result in different drags, as testing will tell.



Nose Cone:

While, primarily, the nosecone is meant to provide a smooth transition from **the leading edge to the body of the car**, in a way which minimises both **drag and weight**. However, we found that it could be a great opportunity to create lift at the front, decreasing the normal force and preventing the **tipping movement** at the front of the car.



Rear Wing:

The rear wing has **no effect on drag**: rather, it can be used to control **lift and therefore downforce**. Ultimately, as a large part of our design philosophy involved around created lift, we decided to sue the rear wings to create this lift, thus balancing out the weight of the gas canister on the back of the car. The only design variation was doing this using the Bernoulli effect, while **minimising extra drag losses**.



All fig.
Initial prototype demonstration All sketches are Hand-drawn by Team

Testing:

The whole page is built on this key:

Strong	Good	Weak	Bad
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In terms of testing methods, virtual testing was intended to calculate intended forces on the car, and visualise flow lines in order to identify aerodynamic weaknesses. However, FEA testing was also done to determine the durability on each component.

- CFD testing – this involved creating a **virtual wing tunnel around our car**, and then pushing air through it at 20m/s to **visualise the flow of air** around the car geometry. We used **Ansys discovery for component testing** to determine flow trends, due to its speed and ease of setup. However, for the actual force values, we ran a **headless fluent simulation due to increased accuracy**.

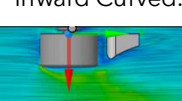
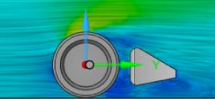
A **custom ML force determiner**: - due to the **speed** of full virtual testing, and lack of appropriate devices within team members, we set up an ML simulation, which took the car geometry and **estimated the force values**, based on curvature and other geometric factors. This allowed for the quick iteration when full testing was not available

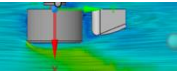
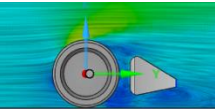
We **FEA tested** (finite element analysis) every component to ensure they could withstand the track conditions, with **20N** applied in them to account for forces well above drag and race launch forces. Throughout this document, we scored component **durability relative to the FEA scores in Fusion 360**, 10 being extremely unlikely to break and 1 being almost certain to break.

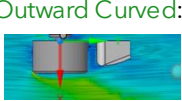
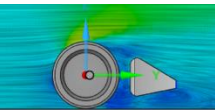
All force values are taken at 20m/s for easy exponential growth to higher or lower speeds. All weights are in newtons.

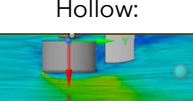
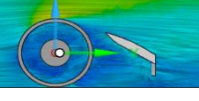
Rear Pods:

The Rear pods were a relatively simple component, because there is little variation in its overall shape: it should fill up the shape of the wake behind the wheel. The only variation **in design is minor refinement and saving weight**, or using it to move the centre of mass.

Drag Co	Weight	FEA Score	Inward Curved:	
0.22	17.2	9		

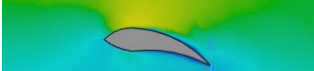
Drag Co	Weight	FEA Score	Straight:	
0.231	17.9	9		

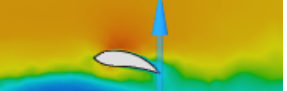
Drag Co	Weight	FEA Score	Outward Curved:	
0.217	19	9		

Drag Co	Weight	FEA Score	Hollow:	
0.231	14	7		

Overall, the outward curved pod was the best shape by drag. However, in order to minimise mass, we explored a hollow option, which resulted in a similar drag but a much lighter weight: both designs were to be explored and used in later prototypes.

Rear Wings:

Pointed:	Drag	Lift	FEA Score	
	0.07	0.08	7	

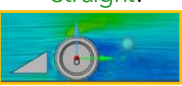
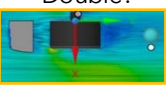
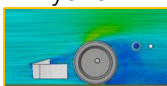
Refined:	Drag	Lift	FEA Score	
	0.05	0.09	7	

A sharper design, with a point as the leading edge rather than a curve seemed to work better aerodynamically, without affecting the amount of lift produced.

Front Wing:

General Shape:

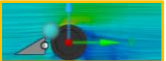
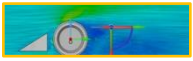
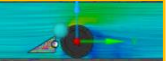
The front area of the car has 2 parts: the nosecone and the wing. The wing itself, as seen before, has two main purposes: to prevent face-on drag of the wheel while reducing the Magnus effect behind it. Below we evaluate different geometries:

Curved:	Straight:	Double:	Hybrid:
			
Legality	Legality	Legality	Legality
Magnus effect	Magnus effect	Magnus effect	Magnus effect
Face-on Drag	Face-on Drag	Face-on Drag	Face-on Drag
Total Drag	Total Drag	Total Drag	Total Drag
Normal Force	Normal Force	Normal Force	Normal Force
FEA score	FEA score	FEA score	FEA score

The overall conclusion was that simpler designs faired better due to better FEA Score and less face-on drag. The biggest problem in designing the front wing was the **indirect proportionality of the magnus effect behind the wheel and wheel face-on drag**. This is because face-on drag is reduced by a steeper angle at which airflow is directed, which amplifies the magnus effect as the wheel would have a larger wake. The most efficient shape was the slight curve, which balanced this angle out.

Refinement:

To refine the wing, we endeavoured to guide more airflow around the wheel, as opposed to over it, which aided all aspects of design – it removed material, allowing it to be lighter; the wheel's wake is minimised from the top and face-on drag is reduced as less air hits the wheel.

Inward Curved:	Outward Curved:	:Straight	Fins:
			
Legality	Legality	Legality	Legality
Magnus effect	Magnus effect	Magnus effect	Magnus effect
Face-on Drag	Face-on Drag	Face-on Drag	Face-on Drag
Total Drag	Total Drag	Total Drag	Total Drag
Normal Force	Normal Force	Normal Force	Normal Force
FEA score	FEA score	FEA score	FEA score

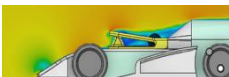
Overall, the outward curved side prevailed as the safest and most efficient option.

Nosecone:

The nosecone is incredibly strategic in terms of preventing the ‘tipping movement’ , producing a reasonable amount of lift and providing a smooth transition between the front of the car and the halo.

Double Sided:	Drag	Weight	Lift	FEA Score	
	0.3	47	.01	9	

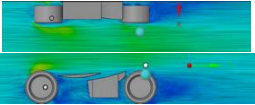
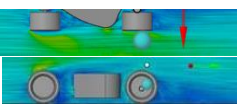
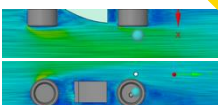
Cone:	Drag	Weight	Lift	FEA Score	
	0.29	45	.008	9	

Fins:	Drag	Weight	Lift	FEA Score	
	0.25	44	.01	8	

We found that the design which directed 22% of airflow below the car (relative, from the bottom of the car to the halo platform) was the best, as it allowed for the correct amount of lift, preventing the car from flying off the track, while not producing more drag than necessary. It also balanced a good angle of attack with the length of the nose cone, which adds unwanted weight. We also moved the wings further up the nosecone, rather than behind it to provide space for an increased wheelbase. Further, we decided to curve the nosecone out, as seen in figure. To prevent airflow hitting the wheel, with 2 fins to amplify this effect below the car.

Side Pods:

The side pods were the most developed component for us, due to their importance in defining the drag of both the front wheel and the rear wheel, and providing a comfortable guidance of frenzied airflow from the halo to the rear of the car.

Fins:	Side Curve:	Full Curve:
		
Legality	Legality	Legality
Front wheel	Front wheel	Front wheel
Rear wheel	Rear wheel	Rear wheel
Total Drag	Total Drag	Total Drag
lift	lift	lift
Airflow stability	Airflow stability	Airflow stability
Weight	Weight	Weight
FEA score	FEA score	FEA score

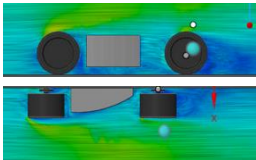
Our first generation of results reenforced an idea found in the front wing development: simpler design are often better. The side curve, the best design, was effective at preventing rear wheel drag, but was weak in aiding the front wing, providing the potential for further development.

2nd Generation:

Reversed.

Legality	
Front wheel	
Rear wheel	
Total Drag	0.2

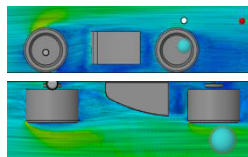
Lift	.004
Airflow stability	
Weight	32.2
FEA score	7



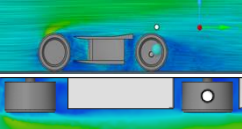
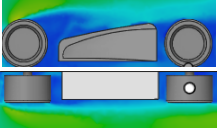
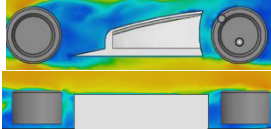
Flat Front:

Legality	
Front wheel	
Rear wheel	
Total Drag	0.18

Lift	.004
Airflow stability	
Weight	30.1
FEA score	7



Our second generation of ideas brought a new light to life: having a flat front, as shown in the previous 2 designs, was extremely effective at preventing front wheel wane. This is because high pressure forms in front of the wall, with low pressure in front of that: balancing out the regions, perhaps giving us a slight boost. A larger wall at the front was better, as shown in reverse.

Flat Front - Fins:	Front Wall	Front Wall Light
		
Legality	Legality	Legality
Front wheel	Front wheel	Front wheel
Rear wheel	Rear wheel	Rear wheel
Total Drag	Total Drag	Total Drag
Lift	Lift	Lift
Airflow stability	Airflow stability	Airflow stability
Weight	Weight	Weight
FEA score	FEA score	FEA score

To conclude our side pod development, we decided have a wide (18mm) but short , as this prevented face-on drag while filling up the wheel wake. Furthermore, we introduced a top fin to guide airflow over the rear wheels effectively, while indenting the side of the pod to save weight. We debated upon the length of the side pod and concluded that a full-car analysis could give us the best course of action, as it primarily impacted weight, wheelbase and COM offset.



Analysis of CFD Softwares:

- Ansys Discovery:**
- Biggest brand in the Simulation market
 - Relatively accurate
 - Fast, good for quick iterating
 - Large community of help, easy interface

- Ansys Fluent:**
- Biggest brand in the Simulation market
 - Most accurate, used by the biggest brands
 - Good API for automation
 - Slow, hard to learn interface
 - Smaller community of help

- Openfoam:**
- All simulations are free
 - Open-source, allowing for customization, integration into apps and automation
 - No GUI, only TUI, making it harder to learn and visualize settings
 - Some simulations have to be custom written
 - Smaller community of help, harder to troubleshoot
 - Available on all OSs
 - Mildly accurate

- Autodesk CFD:**
- Free for students, works well with other Autodesk products
 - Mildly accurate
 - Easier to learn than Ansys
 - Less use-cases, only for fluid dynamics, with less complex physics involved
 - Smaller community of help, but great customer service
 - Only available on windows

- Simscale:**
- Powered by Openfoam
 - Easiest software to use, intuitive GUI
 - Widest range of use cases, from FEA to thermal analysis to CFD
 - Run in browser, can be accessed anywhere on any device
 - Mildly accurate
 - No API for automation

Design Goals:

107mm wheelbase

0.44N drag

13mm COM offset

56g Weight (before paint)

0.15N lift

Final Prototypes:

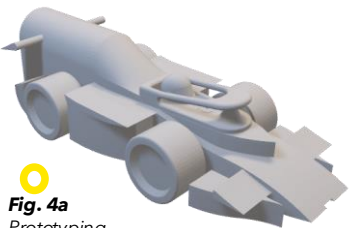


Fig. 4a
Prototyping

Xevioso	
COM Offset:	16.2mm
COM Width:	76mm
Wheelbase:	70mm
Weight:	56.65
Drag (20m/s):	0.4N
Lift (20m/s):	0..27
FEA Score:	6
Race Time:	Breaks down

Overall, Xevioso's major advantage was weight and weight distribution. The side pods are especially light, **successful preventing the Magnus** effect. However, Xevioso was not perfect, but demonstrated 4 disadvantages :

- By producing lift, drag was sacrificed, as flow from under the car rapidly looped around.
- **Low wheelbase** and **high COM - unstable**
- A low FEA score due to fins
- The fit was too **huge**, meaning the car would fly of the track - thus the **model breaks down**.

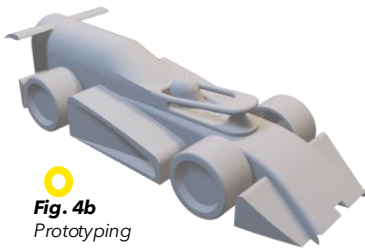


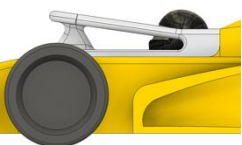
Fig. 4b
Prototyping

Raijin	
COM Offset:	14.5mm
COM Width:	72mm
Wheelbase:	106mm
Weight:	60g
Drag (20m/s):	0.47N
Lift (20m/s):	0.17N
FEA score:	9
Race Time:	1.35

Raijin was much stronger than Xevioso, but was built on the **same ideas of eliminating front** wheel wake, and a gradual push above the rear wheels. It has a far better FEA score, with a **lower centre of mass**, a higher wheelbase, making it more stable. In addition, it prevented airflow from looping around the car **due to plates o the bottom**. We found that, coincidentally, this reduced the size of the car's wake as well. However, it was very heavy, forcing us to adopt hollow sections in the side pods which dramatically increased drag.

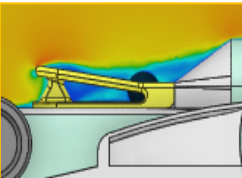
Further Refinement:

Initially, we assumed the 30mm halo platform offset, under the belief that the loft from the gas canister chamber would have a smaller angle of attack and therefore produce less drag. However, this posed 3 problems:

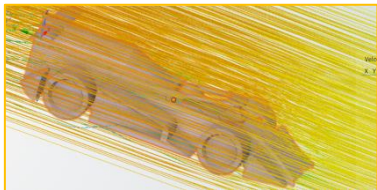


It weighted the most, potentially posing the problem of going above the target of 60g

It due to this **increased weight**, and its position in the top half of the car, the centre of mass was considerably **moved up**, negatively impacting race stability, which is far more **important than drag**.



In terms of legality, having the 30mm offset gave some problems: having a wheelbase below 115mm would have resulted in the wheels **obstructing halo visibility** to the side, because the halo was severing places far forward. We decided that this wheelbase would increase drag far more than reducing the halo platform **offset to 20mm, resulting in us doing so**.



Overall, we found that Raijin beta was a better design, **producing the right amount of lift** and a low drag of.... We found that the plates in front of the main side pod were incredibly effective at **preventing airflow under the car looping around and creating drag, improving stability**. Furthermore, the wheelbase of Xevioso was far too small to even consider.

However, the centre if mass offset, while being in a good place width wise, had a very high offset of 14.7mm. To bring this down, we **introduced side plates on the bottom** of the side pods, at 4mm thickness .



Fig. 4c-f
Prototyping

At this point, we came up with the realistic lift goal of 0.15N lift at 20m/s, to account for the fastest potential car speed of 35m/s, preventing the car from being lifted off the track. We included a safety margin, with the car being able to travel up to **45m/s** without exceeding weight.

Design Tree:

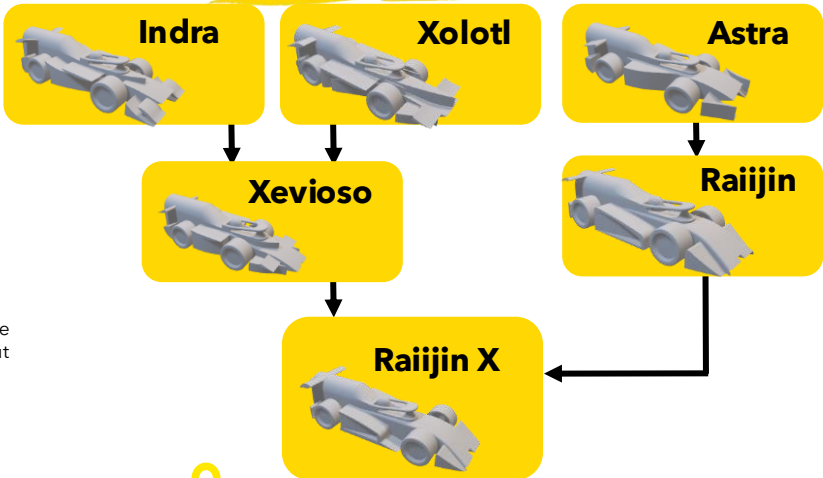


Fig. 4g
Design development tree

Final Design:

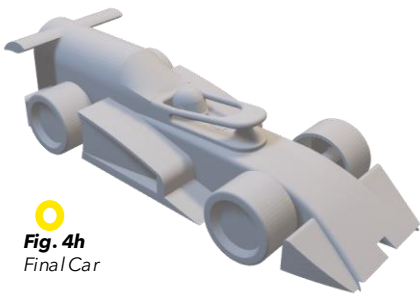


Fig. 4h
Final Car

Raijin X

COM Offset:	14mm
COM Width:	69mm
Wheelbase:	110mm
Weight:	56.65
Drag (20m/s):	0.42N
Lift (20m/s):	0.17N
FEA Score:	8
Race Time:	1.32

FEA Testing:

Overall, there were no major concerns with the durability of the car, as shown in the ansys FEA analysis. However, we did identify **2 weak points that could potentially break in the event of an accident**, such as, for example, the car being dropped. However, the parts do not effect legality, and their loss, while slightly **hindering aerodynamics** would not be too severe. Nevertheless, their loss is unlikely in the first place.

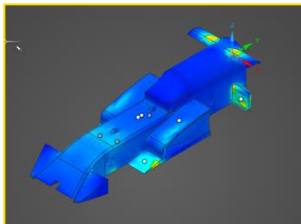


Fig. 4h
Final Car FEA Testing

Manufacturing Prep

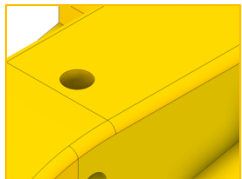


Fig. 4j-l
Preparation - manufacturing

To account for inaccuracy in our school CNC router, we had the take multiple steps:

- We introduced a **1mm tolerance to all components restricted by legality** . This did not include the faces of the no-go zone, due to the aerodynamic implications, but ensured it was accurate after manufacturing.
- Due to the size of the drill bit being **6.35mm**, we needed to increase the size of the axle hole, increasing the axle friction, but ultimately inevitable. Again, due to the 6.35mm drill bit, our router was unable to create sharp edges, and attempting to do so would often result in inaccuracy. To prevent this, we **filleted** the edges of the body.

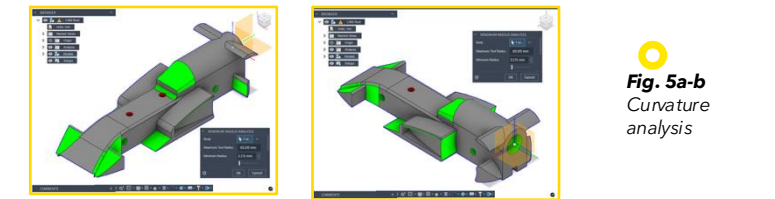
What is CNC Routing / CAM?

At its core, CNC (Computer Numerical Control) routing, Which falls under the umbrella computer aided manufacturing (**CAM**), is a type of CNC machining (pre-programmed software, dictating movement of machinery for manufacturing) used to cut wood, plastic or metal. The CNC router consists of four main parts: the cutting bed, the spindle, the drive system, and the controller. CNC routers can cut along three directions: X-axis, Y-axis, and Z-axis. The router works first by inputting the CAD of the car cut, then switching to the toolpath program for cutting instructions. After this, the material is then secured to the cutting bed, and the correct tools/bits are inserted to cut the material, polyurethane, into the desired shape along the 3 axes.

Preparing for Manufacturing

MRA (Minimum radius analysis):

To make sure that our car is cut **accurately**, we needed to make sure that the router bit that our manufacturing department (in our school) has on the actual router can be used to cut out our car, made of polyurethane. Since the diameter of the router bit is 6.35mm (radius = 3.175mm) , we know that any holes / gaps under this length would be **unfeasible** to cut via router. Therefore, we checked using our CAD software (figure 2 & 2.1) where the manufacturing. The colour coding for the image is as follows::



Grey: Not Applicable to the search - this can easily be cut by the router.

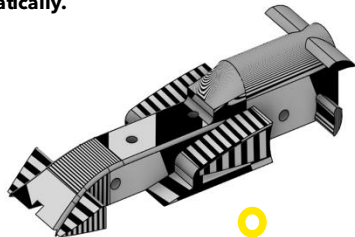
Green: The router can cut this area. However, These areas have curves and so therefore must be checked.

Red: The router can't cut this area, and defects are very likely to happen when this body is being manufactured.

As you can see in the images, all areas can be cut in our final design (except for the halo holes, which we would have to drill in). Using This tool allows us to check every car component's **manufacturability** in a precise and simple manor. To ensure sharp edges could be cut, we **filleted them as much as reasonable** and found that **slicer software would aid in fixing other problems automatically.**

Zebra Analysis:

We conducted a Zebra analysis to detect the curvature of the car. This allowed us to identify **dangerous** areas where curvature **is high**, such as the gas canister-halo loft, as well as see imperfections in design easier. This allowed to detect an **imperfection** in our loft, and clean up the geometry..



Router problems:

When CNC routing, we faced 2 main problems:

G-code Inaccuracies:

When converting our car's 3MF file to g-code, which usually occurs successfully using slicing technology. However, a problem we faced, and many other teams from our school faced was a key inaccuracy in this conversion: a line, **deviating off the centre line of the car, was created, like a groove.** There was no creation of such in our CAD drawings, not any problems in exporting, leading us to believe it was a **slicer problem.** However, due to our lack of expertise in g-code, we concluded that it would be more efficient for us to simply sand it down, carefully and considerate of legality, than spend tens of hours researching how to fix such a problem. However, the **final car's dimensions would be identical.**

CNC router bit Simplification:

We found that, due to our school of outer drill bit being 6.35mm, certain **parts edges could not be cut due to their sharpness.** However, these were not shown in our analysis due to the fact that the slicer software, by default, fixed these sharp edges by making them curved. We found that this did not have any **major legal or aerodynamic effects,** and we simply had to monitor default changes to design. Most change occurs around the bottom the rear wings and the edges of the rear pods. The only significant problem it would develop is the potential need to produce add-on parts to the wings, due to breakage of manufacturing error, as our CAD drawings would be inaccurate in comparison to the real car.



Fig. 5c
Zebra analysis

Fig. 5d
CNC router

Risk Assessment :

Risk	Cause	Solution
Cuts/ Scrapes	Use of saws and sandpaper.	Use Safety techniques like , mark out cutting lines
Eye Irritation	Sawdust exposure for long period of time. Paint blown on to eyes.	Increase d ventilation, do not paint in windy conditions.
Skin Irritation	Sawdust exposure for long period of time. Paint contact with skin.	Increase d ventilation, wear an apron when painting.
Clothes caught	Loose Clothing around machinery.	Wear an apron, tuck in loose clothing, cover machinery.

CNC aftermath:

Once the car was cut, we had to do multiple things to ensure the surface for painting was accurate and loyal to our CAD geometry.

Removing front block. As mentioned before, we had to leave a front block in front of the car due to the increased length of the model block compared to our car. Our school DT **teacher removed this for us using due to restrictions on age at our school, though we draw the paint at which to cut.**



Sanding - to remove the **deformities** in the CNC process, we had to sand down the car. Thai was a tricky process due to the consideration of breaching the no-go-zone and legality. At the beginning, we carefully used **150 grit sandpaper, but in the final stages we used 2500 grit sandpaper.**



Before even beginning the next phases, we tested the car to ensure no significant errors were made. On our first cut, there was a slicer error with removed 1mm of material from the bottom of the car, which demonstrated to need to conduct such tests. We measured key dimensions and tested durability of weak parts to ensure a high quality cut.



Fig. 5e-g

Component Preparation:

Polishing:

We polished many parts of our design such as:

3D printed parts such as the halo, helmet and the wheels needed to be polished to smooth out layers formed in **FDM without changing their geometry.**



Axles, in order to **reduce rolling resistance** by removing waste material, while remaining legal by not using lubricant. It also means that it will slot through the axle bushes with ease, making for easy assembly.

Cutting axles:

Another crucial adjustment we had to make during the assembly process was modifying the length of the axles. Initially, our axles were designed to be **66 mm long.** However, when we conducted an assembly test, we quickly noticed an issue: the wheels were not staying securely in place. Instead, they were sliding side to side along the axle, which indicated that the axles were too long for our intended design. Leading to a **potential fault & inefficient aerodynamics**



Fig. 5h

To resolve this problem, we decided to shorten the axles. Our first adjustment involved cutting them down by 7 mm, bringing the length to 59 mm. However, after conducting another test, we realised that the axles were still slightly too long, allowing for some unwanted movement in the wheels. To refine the fit further, we made an additional cut, shortening the axles by another 2 mm. This final adjustment brought the axle length to 57 mm, which proved to be the optimal size.

Surface Finish:

Process	Evaluation
Black Primer	We Used 3 coats of primer to provide a base for later finishing. Primer is used as an initial coat in most things because it allows later coats of paint and gloss to stick to the car, which will allow a smoother coat of gloss later.
Black Gloss	We decided to skip the paint and go straight to a gloss finish, because the primer achieved the same black coat that paint would. We applied one layer of gloss due to lack of need to add any extra weight. The gloss created a shiny finish to enhance aesthetics, with no real aerodynamic benefits.
Decals	The decals we used were printed in viny sticker, due to their cheapness and ease of use, and also the lack of time to prepare the panned water-based decals. While they achieved their purpose overall, by branding our car and providing sponsor return on investment, they created an extruding edge by fractions of a millimetre, which forced us to apply a layer of lacquer.
Lacquer	Lacquer was applied in order to get rid of the effects of this extrusion and provide a neat and clean finish overall without harming aesthetics.
Painting Jig	We used a custom-made painting jig to hold our car by the gas canister chamber to paint easily.



Fig. 5i-l

Quality Control:

Sanding
After every layer of finish, we **sanded** down the car to ensure a smooth surface was maintained, and to ensure no deformities such as blotches of paint were preserved in the final car.

Scrutineering
Once the final car was completed, we measured all measurements **restricted by legality.** This ensured no mistakes could have happened during sanding.



Fig. 5m-n

Evaluation:

Process	Evaluation
Front wing	Aero dynamically successful: but if CNC router limitations could be overcome design could be improved through a dded complexity.
Side Pods	Aero dynamically successful: If weight could be preserved in other areas and place here design could be made better. Allows for a great wheelbase and COM offset.
Rear Pods	Aero dynamically successful: no real improvement, other than potentially increasing length with it sacrificing wheelbase or weight.
Rear Wing	Aero dynamically sufficient: produces a considerable wake without targeted lift production.
Nose cone	Aero dynamically successful- produces lift without sacrificing drag. Could be more durable, could be a weak point.
CNC routing	Not successful- numerous deformities and minor design flaws, had to fix- could be better with better CNC router.
Sanding	Successful – smooth painting surface despite CNC deformation .
Painting	Sufficiently successful – coats were applied well without blotching, but better-quality paints, paired with a better cut could have resulted in a cleaner finish.
Decals	Overall livery is good, sponsors communicated well – water slide would have been preferable, but lack of time lead us to point vinyl
Finishing	Sufficient - finish was a decent quality, with . Good livery but could have been cleaner if more time had been made available.

